

Observer, Platonic Representation Embedding Spaces

Conversation summary and direct implications for the 7-representation-space experiment

Core idea

Different models can use different coordinate systems while preserving part of the same latent relational structure. The key question is therefore not whether the vectors themselves are similar, but at what level geometry is preserved: local neighbors, distance rankings, a linear subspace, or even the same geometry up to rotation/reflection.

1. The Three-Dimensional Time Model and the Observer

In the model discussed, the Observer is not an additional time dimension. It is the observation/integration window moving within a space described through three aspects of time:

- Objective Time - physical/clock time: the sequence in which events are realized and measured.
- Subjective Time - the rate and duration of the Observer's experience. The same objective interval can feel shorter or longer.
- Alternative Time - the space of possibilities available from the current state: different branches that could occur.

$$\text{Observer}(t) = (T_{\text{objective}}, T_{\text{subjective}}, T_{\text{alternative}})$$

NOW is the meeting point: one possible future becomes an actual event and enters the objective trajectory while also being experienced by the Observer.

2. Nested Observer Windows

Nested Observer Windows add a hierarchy: small, fast systems process information over short windows, while larger systems integrate information over longer windows.

neurons -> local circuits -> brain regions -> networks -> APEX observer

Each level can be viewed as a distinct integration window. The APEX observer is analogous to the unified experience of "me now." NOW therefore need not be a mathematically dimensionless instant; it can be an integration window with duration.

3. Connection to the Platonic Idea

The Platonic connection is a useful philosophical interpretation, not an identity proven by the time model. A Platonic space can be thought of as a space of structures or possibilities, with experienced reality as one trajectory through that space.

possibility -> selection/realization -> actuality -> experience

In an ML analogy, Platonic space can be viewed as a shared latent manifold, while each encoder provides a different chart or coordinate system for that latent structure.

$$f_i : X \rightarrow Z_i$$

Here X is the latent structure of the data and Z_i is the representation space of encoder i . Z_i and Z_j do not need to have the same dimensionality or coordinates in order to preserve similar relations.

4. Direct Connection to Your Experiment

The experiment contains 7 representation spaces over the same 9,533 items. This is especially powerful because the rows are aligned: the same sample occurs in the same row in every representation.

Space	Representation matrix
img_small	9,533 x 768
img_base	9,533 x 1,536
img_large	9,533 x 2,048
txt_bge	9,533 x 1,024
txt_gpt2	9,533 x 768
txt_bert	9,533 x 768
txt_sbert	9,533 x 768

For example, if A is `img_small`:

$$A \in \mathbb{R}^{(9533 \times 768)}, \quad a_i \in \mathbb{R}^{768}$$

Each row a_i is the embedding of one sample. If B is `img_base`, then B has shape 9,533 x 1,536, and row i in A must correspond to the same underlying sample as row i in B .

5. Why 21 Pairs but 42 Mapping Directions?

For a symmetric structural-similarity measure such as CKA or neighborhood overlap, A versus B is equivalent to B versus A . With 7 spaces there are:

$$C(7,2) = 7*6/2 = 21 \text{ unique pairs}$$

Mapping, however, is directional. $A \rightarrow B$ need not perform as well as $B \rightarrow A$, so there are:

$$7*6 = 42 \text{ directed mappings}$$

This asymmetry can itself be scientifically informative. If $A \rightarrow B$ is substantially stronger than $B \rightarrow A$, one representation may contain richer information, or the information in the other may be more simply recoverable from it.

6. Hypothesis: Convergence Toward Shared Relational Geometry

The central intuition is that sufficiently capable models trained on the same underlying reality may converge toward a shared relational structure. The precise claim is not “the same coordinates,” but shared relational geometry.

same underlying world -> different encoders -> partially shared geometry

For example, if dog is close to wolf and far from airplane in one space, another model that captures the same semantics well may preserve at least that ordering even if distances, angles, and dimensionality differ.

Important distinction

Local geometric agreement is a weaker and more plausible claim than global metric identity. Two models can share the same local neighbors while warping global space in different ways.

7. Hierarchy of Tests - From Weakest to Strongest

Hypothesis	Meaning	Fast test
H1: Shared neighborhoods	The same samples are neighbors of a sample in both spaces.	kNN overlap / RBO-like neighborhood agreement
H2: Shared ranking	The ordering of distances is similar even if their values differ.	Spearman over pairwise distances / ranks
H3: Shared global structure	Global structure is similar under a suitable invariant measure.	CKA / RSA
H4: Shared linear subspace	Shared directions exist that can be linearly aligned.	CCA / Ridge
H5: Same geometry, different coordinates	The spaces are nearly identical up to rotation/reflection.	Orthogonal Procrustes

8. Fastest First Test: kNN Overlap

For every sample i , find its k nearest neighbors in each space:

$$N_A^{k(i)}, \quad N_B^{k(i)}$$

Then compute the overlap fraction:

$$\text{Overlap}_k(i) = |N_A^{k(i)} \cap N_B^{k(i)}| / k$$

Average over all samples. Rather than using a single k , test a range of scales:

$$k \text{ in } \{1, 5, 10, 20, 50, 100, 500\}$$

This reveals whether the common geometry is only local or remains stable over increasingly broad neighborhoods.

9. CKA - Global Structural Similarity

CKA is especially useful when representation dimensions differ, making it suitable for the 7 spaces. It gives a symmetric measure of global structural alignment, but does not prove that embeddings can actually be translated between spaces.

$$\text{CKA}(X,Y) = ||X^T Y||_F^2 / (||X^T X||_F ||Y^T Y||_F)$$

CKA is therefore useful as a candidate filter and a map of structural similarity, but a mapping test is needed to answer the exportability question.

10. Mapping Test: Can an Embedding Be Exported?

If A is the source and B the target, learn a transformation on the training set only:

$$B_{\text{hat}} = A W$$

For example, for img_small -> img_base:

$$A_{\text{train}}: 8533 \times 768, \quad B_{\text{train}}: 8533 \times 1536, \quad W: 768 \times 1536$$

On evaluation data compute $B_{\text{hat_eval}} = A_{\text{eval}} W$ and compare it with the true B_{eval} . Retrieval is preferable: does the predicted vector for sample i retrieve the true target vector of the same sample among its nearest neighbors?

$$\text{metrics: } R@1, R@5, R@10$$

11. Why PCA + Procrustes Is a Clean and Fast Test

Because of the sample-to-dimension ratio, especially for img_large, a full dense mapping contains millions of parameters and can be less clean experimentally. A compact funnel is therefore preferable:

$$A \rightarrow \text{PCA}_{256} \rightarrow \text{Orthogonal Procrustes} \rightarrow B_{\text{PCA256}} \rightarrow \text{retrieval}$$

Orthogonal Procrustes searches only for a rotation/reflection:

$$Y \sim X R, \quad R^T R = I$$

If such a constrained mapping performs well on evaluation data, that is strong evidence that the spaces share nearly the same geometry and mainly differ in coordinate system.

If Procrustes is weak but Ridge/CCA is strong, there may be shared linear structure without isometry. If kNN is strong but linear mapping is weak, the spaces may share a manifold with a nonlinear warp.

12. CCA - Is There a Shared Latent Subspace?

CCA searches for directions in the two spaces whose representations are maximally correlated:

$$u = X a, \quad v = Y b, \quad \text{maximize } \text{corr}(u,v)$$

A long sequence of high canonical correlations suggests a shared linear latent subspace. This is useful when the two spaces share substantial information while also retaining private information.

13. Compatibility Matrix and the HUB

Instead of choosing a HUB by dimensionality alone, define it from measured cross-space compatibility. Build several complementary matrices:

- M_{CKA} - global structural similarity.
- M_{kNN} - preservation of neighborhood structure.
- M_{CCA} - shared linear subspace.
- $M_{translate}$ - $R@K$ for every source $\rightarrow$ target direction.

The translation matrix is directional and need not be symmetric. A good HUB is a space from which other spaces are easy to reach, or under a stronger criterion, a space with strong bidirectional compatibility.

$$H^* = \operatorname{argmax}_H \operatorname{mean}_j \operatorname{Compatibility}(H, j)$$

In this sense the HUB is a kind of geometric medoid of the representation family, not necessarily the largest or highest-dimensional space.

14. Interpreting Possible Results

Observed pattern	Interpretation
CKA high + Procrustes high	Nearly the same geometry, different coordinates.
CKA high + Ridge high	Shared structure that can be linearly aligned.
CCA high + CKA moderate	A shared subspace plus private information in each encoder.
kNN high + linear map weak	Similar topology/manifold with nonlinear warping.
A $\rightarrow$ B strong, B $\rightarrow$ A weak	Asymmetry; possible information containment or differences in representational richness.
CKA low + kNN low	Representations are likely substantially different for the tested data.

15. Does Local Similarity Rule Out Global Regularity?

No. Local similarity alone does not prove that there is no single global regularity. It only says that the strongest empirical evidence may exist at the local level: models agree on what is near what even if they do not agree on global distances, angles, or coordinates.

shared local relations $\neq$ identical global metric

Imagine one latent manifold M and two different encoders:

$$f_A : M \rightarrow Z_A, \quad f_B : M \rightarrow Z_B$$

The two maps can apply different nonlinear warps while still preserving neighborhood topology. A different global metric is therefore not equivalent to the absence of common structure.

Map analogy

Two maps of Earth can use very different projections: global distances and areas change, while local adjacency is often preserved. Likewise, two encoders may be different charts of

the same relational manifold.

16. Two Competing Hypotheses: Model-Induced vs Universal Structure

To interpret shared neighborhoods, distinguish two possible explanations:

- Weak universality / model-induced similarity - similarity mainly comes from something already shared: the same modality, a similar architecture, a similar objective, or similar training data.
- Platonic / universal relational structure - even models with different modalities, architectures, and training regimes converge on part of the same relations because the world itself has stable structure.

Similarity = world-induced component + model/training-induced component

If only within-family pairs show high agreement, architecture/modality becomes the stronger explanation. If substantial agreement survives across modality and architecture boundaries, it becomes harder to explain using model similarity alone.

17. How to Test Whether Attention or Modality Explains Similarity

For each of the 21 pairs, attach metadata and compare groups:

- Same vs different modality.
- Same vs different architecture family, for example Transformer/attention versus another architecture.
- Same vs different training objective.
- Same vs different model family / pretraining source.

$E[S \mid \text{same modality}]$ vs $E[S \mid \text{different modality}]$

$E[S \mid \text{same architecture}]$ vs $E[S \mid \text{different architecture}]$

S can be kNN overlap, rank agreement, CKA, or translation score. With only 21 pairs, this should initially be treated as exploratory analysis rather than strong statistical proof; if possible, extend the experiment with additional models.

Observed pattern	What it suggests	Interpretation
Only within-family similarity is high	Geometry depends strongly on modality/architecture/training.	Model-induced explanation strengthens.
Cross-modal similarity rises with model quality	Better models converge toward shared relations despite modality differences.	Platonic/universal explanation strengthens.
kNN high but CKA/Procrustes low across modalities	Shared neighborhoods but different global metric.	Shared topology + different geometry.
Cross-architecture + cross-modal agreement remains high	Similarity is not easily explained by attention or family alone.	World-induced relational structure becomes more plausible.

18. A More Precise Platonic Formulation

Instead of claiming that all models converge to one identical embedding space, a more flexible and precise formulation is that a partially shared relational manifold exists and every encoder builds its own chart of it:

$$\text{exists } M: \quad Z_A = f_A(M), \quad Z_B = f_B(M)$$

The hypothesis does not require $Z_A \sim Z_B$ globally. It asks which invariants survive across charts: neighborhood, ordering, topology, subspaces, or near-isometry.

Expanded research question

Is representational similarity explained mainly by shared architecture/modality, or does a common local relational geometry survive across those boundaries?

19. Recommended Next Experiment

For a fast, inexpensive, and clear answer, use this sequence:

- 1) Compute kNN overlap for all 21 pairs at several k values.
- 2) Compare with the existing CKA matrix to separate local from global similarity.
- 3) For all 42 directions: reduce both sides to the same dimension, e.g. PCA-256, then fit Orthogonal Procrustes.
- 4) On the 1,000 evaluation samples, report only R@1, R@5, and R@10 - with no tuning on evaluation data.
- 5) For interesting pairs where Procrustes is insufficient, add Ridge/CCA and only then consider a nonlinear adapter.

Precise research question

Are different encoders different coordinate charts of partially shared latent manifolds, and if so, at what level is structure shared: local neighborhoods, global relations, a linear subspace, or near-isometric geometry?

20. One-Sentence Summary

The connection between Observer/Platonic ideas and your experiment is conceptual but testable: models may not learn the same vectors, but rather different charts of a partially shared relational manifold. The key test is whether local or global invariants survive across boundaries of modality, architecture, and training objective, using neighborhood overlap, CKA, CCA, Procrustes, and directional retrieval.